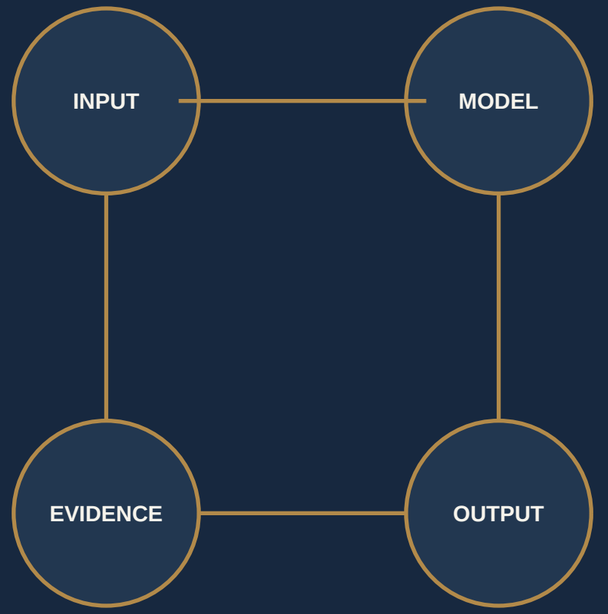
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Internet of Things

AI.201 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define iot architecture.
- Apply key methods including sensors.
- Build or analyze a practical system involving actuators.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

IoT architecture

02

METHODS

Sensors

03

SYSTEMS

Actuators

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · IoT architecture
IoT architecture

02 · Sensors
Sensors

03 · Actuators
Actuators

04 · Embedded devices
Embedded devices

05 · Connectivity
Connectivity

06 · MQTT
MQTT

07 · HTTP
HTTP

08 · Cloud systems
Cloud systems

Core ideas, made explicit

AI.201 • Distinguish definitions, assumptions and evidence

- IoT architecture
- Sensors
- Actuators
- Embedded devices
- Connectivity

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.201 • Distinguish definitions, assumptions and evidence

- MQTT
- HTTP
- Cloud systems
- Edge computing
- Security

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
IoT architecture

INPUTS
Sensors

METHOD
Actuators

OUTPUT
Embedded devices

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving iot architecture.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using iot architecture.
- Implement or simulate a baseline using sensors.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.